

Udacity Programming Assignment 4 // Angela Gross // Parallel Programming

U Classroom - Udacity

https://www.udacity.com/course/viewer#!/c-cs344/l-95446457/e-97045331/m-97035154

Apps

Lesson 4 - Fundamental GPU Algorithms (Applications of Sort and Scan) > Problem Set 4

DASHBOARD

CLASSROOM

MATERIALS

DISCUSSION

OVERVIEW

student_func.cu

reference_calc.cpp

reference_hw4.cu

reference_main.cpp

```
1 #include "reference_calc.cpp"
2 #include "utils.h"
3 #include <math.h>
4
5 // =====
6 // student_func.cu
7 // =====
8 // Angela Gross
9 // CSCI-595: Parallel Programming
10 // Homework 4
11 // Spring 2015
12 // =====
13 // This Udacity assignment implements red eye removal, and this is accomplished by first creating a score for
14 // pixel that tells us how likely it is to be a red eye pixel. This is already done in the grading framework
15 // and the function "your_sort" is receiving scores and pixel positions and the scores need to be sorted in
16 // ascending order so the framework knows what pixels to alter to remove the red eye.
17 //
18 // In order to sort these pixels, this code implements a 1-bit radix sort inspired by the following nvidia co
19 // http://www.compsci.hunter.cuny.edu/~sweiss/course_materials/csci360/lecture_notes/radix_sort_cuda.cc
20 //
21 // Since it is only using 1-bit, we only need to scan the items and scatter them (and not use a histogram).
22 //
23 // The somewhat efficient multi-block scan was inspired from the following website:
24
```

Reset

Test Run

Review Instructions

Submit

Continue

Good job!. Your image matched perfectly to the reference image.

Your program ran and executed in 9.900384 ms.

Correct!

Please see detailed feedback in the output area below.

Get Help

[MAD About Problem 4---Extremely Large Number!!! Mar 10](#)

[HW4 scoreboard \(running times\) Feb 2](#)

[Strange Encounter with Parameter Jan 27](#)

[HW4: "Difference at pos 285653 exceeds tolerance of 5" but there is only 220480 positions Jan 11](#)

[The server encountered an error. Please try running again. Sep 1](#)

Start a Discussion

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